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SCHOOL OF ENGINEERING AND TECHNOLOGY
DEPARTMENT OF COMPUTER SCIENCE ENGINEERING

Academic Year: 2025-26	
Semester: III	**Course Code:** CSAIML304
Name of the Course: Introduction to Machine Learning	**Tutorials/week:** 0
Teaching Scheme: TH: 04 Hours/Week	
Examination Scheme: **Formative Assessment:** 50 Marks **Summative Assessment:** 100 Marks	**Credit:** 04

Question Bank

CO-Course Outcome, BL- Bloom's Level (L1, L2, L3, L3, L4, L5, L6)

S.No.	Unit	Question	Marks	CO	BL	Asked in Competitive Exam
UNIT-1						
1	1	What is the primary goal of machine learning? A) **Predict outcomes** B) Store data C) Optimize hardware D) Design algorithms	1	CO1	L1	Infosys Certification 2023
2	1	Which of the following defines learning in machine learning? A) **Acquiring knowledge from data** B) Executing predefined rules C) Storing large datasets D) Reducing memory usage	1	CO1	L1	TCS NQT 2022
3	1	What is the purpose of a validation set in machine learning? A) Model training B) **Evaluate model performance** C) Store test data D) Optimize algorithms	1	CO1	L2	UGC-NET 2022
4	1	Which technique is used to assess model generalization? A) **Cross-validation** B) Data normalization C) Feature selection D) Dataset splitting	1	CO1	L2	TCS NQT 2022
5	1	What is the time complexity of evaluating a model on a dataset of size n?	1	CO1	L1	GATE

		A) O(1) B) **O(n)** C) O(n²) D) O(log n)				2023
6	1	Which of the following is a feature set in machine learning? A) **Numeric attributes** B) Raw data C) Labels D) Weights	1	CO1	L1	UGC-NET 2022
7	1	What is the purpose of the train-test split in machine learning? A) **Evaluate model performance** B) Train the model C) Tune hyperparameters D) Store data	1	CO1	L1	Infosys Certification 2023
8	1	Which cross-validation method uses k subsets of data? A) **K-fold cross-validation** B) Leave-one-out C) Stratified sampling D) Holdout method	1	CO1	L2	TCS NQT 2022
9	1	What is the space complexity of storing a dataset with n samples and m features? A) O(1) B) O(n) C) **O(n*m)** D) O(m)	1	CO1	L1	UGC-NET 2022
10	1	Which dataset is used to tune model hyperparameters? A) Training set B) **Validation set** C) Test set D) Entire dataset	1	CO1	L2	GATE 2023
11	1	What is the main advantage of cross-validation? A) **Reduces overfitting** B) Increases training speed C) Reduces dataset size D) Eliminates noise	1	CO1	L1	Infosys Certification 2023
12	1	What is the time complexity of k-fold cross-validation for k folds and n samples? A) **O**(k*n) B) O(n) C) O(n²) D) O(k)	1	CO1	L3	UGC-NET 2022
13	1	Which of the following is a step in dataset preprocessing? A) **Feature scaling** B) Model training C) Prediction	1	CO1	L1	Infosys Certification 2023

		D) Validation				
14	1	What is the purpose of feature engineering in machine learning? A) **Create meaningful features** B) Train models C) Evaluate performance D) Store data	1	CO1	L3	TCS NQT 2022
15	1	Which technique splits data into training, validation, and test sets? A) **Dataset division** B) Cross-validation C) Feature selection D) Data normalization	1	CO1	L1	GATE 2023
16	1	What is the time complexity of computing summary statistics for a dataset of size n? A) O(1) B) **O(n)** C) O(n^2) D) O(log n)	1	CO1	L1	TCS NQT 2022
17	1	Which of the following is a type of feature in machine learning? A) Categorical B) Numeric C) **Both categorical and numeric** D) None of the above	1	CO1	L2	Infosys Certification 2023
18	1	What is the purpose of a test set in machine learning? A) **Evaluate final model performance** B) Train the model C) Tune hyperparameters D) Store data	1	CO1	L1	UGC-NET 2022
19	1	What is the space complexity of k-fold cross-validation for k folds? A) O(1) B) **O**(k) C) O(n) D) O(n*k)	1	CO1	L1	Infosys Certification 2023
20	1	Which of the following is an evaluation technique in machine learning? A) **Confusion matrix** B) Feature scaling C) Data splitting D) Model training	1	CO1	L4	GATE 2023
21	1	Define machine learning and provide an example.	5	CO1	L1	

22	1	What is the purpose of a validation set in machine learning?	5	CO1	L2	
23	1	Explain the concept of cross-validation.	5	CO1	L2	
24	1	What is the time complexity of evaluating a model on a dataset?	5	CO1	L2	
25	1	Define feature engineering with an example.	5	CO1	L2	
26	1	What is the role of a test set in machine learning?	5	CO1	L4	
27	1	Explain the concept of dataset division in machine learning.	5	CO1	L2	
28	1	What is the space complexity of storing a dataset?	5	CO1	L4	
29	1	Define supervised learning with an example.	5	CO1	L3	
30	1	What is the time complexity of k-fold cross-validation?	5	CO1	L1	
31	1	Explain the concept of feature sets in machine learning.	5	CO1	L1	
32	1	What is the purpose of train-test splitting?	5	CO1	L3	
33	1	What is the space complexity of k-fold cross-validation?	5	CO1	L2	
34	1	Define unsupervised learning with an example.	5	CO1	L2	
35	1	What is the time complexity of computing summary statistics?	5	CO1	L3	
36	1	Explain the history and evolution of machine learning with examples.	10	CO1	L2	
37	1	Discuss the role of dataset division in machine learning with examples.	10	CO1	L3	
38	1	Describe the implementation of k-fold cross-validation with a real-world example.	10	CO1	L4	
39	1	Explain the time and space complexity of evaluating a machine learning model.	10	CO1	L4	
40	1	Discuss the process of feature engineering with real-world applications.	10	CO1	L5	
S.No.	**Unit**	**Question**	**Marks**	**CO**	**BL**	**Asked in Competitive Exam**
UNIT- 2						
1	2	What is the purpose of stratified k-fold cross-validation?	1	CO2	L1	TCS

		A) **Balance class distribution** B) Increase training speed C) Reduce dataset size D) Eliminate noise				NQT 2022
2	2	Which dataset is used to compute summary statistics? A) **Entire dataset** B) Training set C) Validation set D) Test set	1	CO2	L1	Infosys Certification 2023
3	2	What is the time complexity of feature scaling for a dataset with n samples? A) O(1) B) **O(n)** C) O(n^2) D) O(log n)	1	CO2	L2	UGC-NET 2022
4	2	Which of the following is a preprocessing technique? A) **Normalization** B) Clustering C) Classification D) Regression	1	CO2	L1	Infosys Certification 2023
5	2	What is the main disadvantage of leave-one-out cross-validation? A) **High computational cost** B) Fast execution C) Low accuracy D) No validation	1	CO2	L2	GATE 2023
6	2	What is the space complexity of storing a feature-engineered dataset? A) O(1) B) **O(n*m)** C) O(n) D) O(m)	1	CO2	L1	UGC-NET 2022
7	2	Which of the following ensures unbiased model evaluation? A) Holdout method B) **Cross-validation** C) Dataset splitting D) Feature selection	1	CO2	L2	TCS NQT 2022
8	2	What is the purpose of feature selection in machine learning? A) **Reduce irrelevant features** B) Train models C) Evaluate performance D) Store data	1	CO2	L2	Infosys Certification 2023
9	2	What is the time complexity of splitting a dataset of size n? A) O(1) B) **O(n)**	1	CO2	L3	UGC-NET

		C) $O(n^2)$ D) O(log n)				2022
10	2	Which of the following is a benefit of dataset division? A) **Improves model generalization** B) Reduces training time C) Increases dataset size D) Eliminates noise	1	CO2	L2	GATE 2023
11	2	What is the main purpose of a machine learning model? A) **Predict outcomes** B) Store data C) Optimize hardware D) Design algorithms	1	CO2	L2	TCS NQT 2022
12	2	Which cross-validation method is suitable for imbalanced datasets? A) **Stratified k-fold** B) Standard k-fold C) Leave-one-out D) Holdout method	1	CO2	L1	UGC-NET 2022
13	2	What is the time complexity of computing mean and variance for n samples? A) O(1) B) **O(n)** C) $O(n^2)$ D) O(log n)	1	CO2	L3	TCS NQT 2022
14	2	Which of the following is a feature engineering technique? A) **One-hot encoding** B) Clustering C) Classification D) Regression	1	CO2	L2	Infosys Certification 2023
15	2	What is the space complexity of a normalized dataset with n samples and m features? A) O(1) B) **O(n*m)** C) O(n) D) O(m)	1	CO2	L1	GATE 2023
16	2	Which dataset is used for final model evaluation? A) **Test set** B) Training set C) Validation set D) Entire dataset	1	CO2	L2	Infosys Certification 2023
17	2	What is the purpose of data preprocessing in machine learning? A) **Improve data quality** B) Train models C) Evaluate performance D) Store data	1	CO2	L2	TCS NQT 2022

18	2	What is the time complexity of applying one-hot encoding to n samples? A) O(1) B) **O(n)** C) O(n²) D) O(log n)	1	CO2	L1	UGC-NET 2022
19	2	Which of the following is a dataset handling technique? A) **Cross-validation** B) Clustering C) Classification D) Regression	1	CO2	L2	TCS NQT 2022
20	2	What is the main advantage of feature engineering? A) **Improves model performance** B) Reduces training time C) Increases dataset size D) Eliminates noise	1	CO2	L2	GATE 2023
21	2	Explain the concept of a machine learning pipeline.	5	CO2	L1	
22	2	What is the space complexity of a feature set?	5	CO2	L2	
23	2	Define reinforcement learning with an example.	5	CO2	L1	
24	2	What is the time complexity of preprocessing a dataset?	5	CO2	L1	
25	2	Explain the concept of real-life applications of machine learning.	5	CO2	L2	
26	2	What is the purpose of stratified k-fold cross-validation?	5	CO2	L2	
27	2	Explain the role of feature scaling in dataset preprocessing.	5	CO2	L2	
28	2	What is the time complexity of applying one-hot encoding?	5	CO2	L3	
29	2	Define data normalization with an example.	5	CO2	L2	
30	2	What is the space complexity of a normalized dataset?	5	CO2	L2	
31	2	Explain the concept of feature selection in machine learning.	5	CO2	L3	
32	2	What is the time complexity of splitting a dataset?	5	CO2	L3	
33	2	Define the holdout method in machine learning.	5	CO2	L3	
34	2	What is the space complexity of storing summary statistics?	5	CO2	L2	
35	2	Explain the concept of data preprocessing in machine learning.	5	CO2	L4	
36	2	Discuss the role of reinforcement learning in real-world	10	CO2	L4	

		applications.				
37	2	Describe the time and space complexity of feature scaling in preprocessing.	10	CO2	L5	
38	2	Explain the implementation of stratified k-fold cross-validation with an example.	10	CO2	L4	
39	2	Discuss the role of data normalization in improving model performance.	10	CO2	L5	
40	2	Describe the applications of feature engineering in real-world scenarios.	10	CO2	L6	
S.No.	**Unit**	**Question**	**Marks**	**CO**	**BL**	**Asked in Competitive Exam**
		UNIT-3				
1	3	Which of the following is an application of machine learning? A) **Image recognition** B) Database storage C) Hardware optimization D) Network configuration	1	CO3	L1	Infosys Certification 2023
2	3	What is the time complexity of training a model on a dataset of size n? A) O(1) B) O(n) C) O(n²) D) **Depends on the algorithm**	1	CO3	L1	TCS NQT 2022
3	3	Which learning paradigm involves labeled data? A) **Supervised learning** B) Unsupervised learning C) Reinforcement learning D) Semi-supervised learning	1	CO3	L1	UGC-NET 2022
4	3	What is the main process in machine learning? A) **Model training** B) Data storage C) Hardware optimization D) Algorithm design	1	CO3	L1	TCS NQT 2022
5	3	Which of the following is a real-life example of machine learning? A) **Spam email filtering** B) Manual data entry C) Network routing D) Hardware design	1	CO3	L1	GATE 2023

6	3	What is the space complexity of storing a trained model? A) O(1) B) O(n) C) O(m) D) **Depends on the model**	1	CO3	L2	UGC-NET 2022
7	3	Which learning paradigm uses rewards and penalties? A) Supervised learning B) Unsupervised learning C) **Reinforcement learning** D) Semi-supervised learning	1	CO3	L2	Infosys Certification 2023
8	3	What is the main advantage of supervised learning? A) **Labeled data availability** B) Fast computation C) No need for data D) Low memory usage	1	CO3	L2	TCS NQT 2022
9	3	What is the time complexity of predicting with a trained model? A) O(1) B) O(n) C) O(n^2) D) **Depends on the model**	1	CO3	L3	UGC-NET 2022
10	3	Which of the following is a reinforcement learning application? A) **Robot navigation** B) Email filtering C) Image classification D) Data clustering	1	CO3	L3	GATE 2023
11	3	What is the purpose of the machine learning pipeline? A) **Automate model development** B) Store data C) Optimize hardware D) Design algorithms	1	CO3	L2	Infosys Certification 2023
12	3	What is the time complexity of preprocessing a dataset of size n? A) O(1) B) **O(n)** C) O(n^2) D) O(log n)	1	CO3	L1	UGC-NET 2022
13	3	Which learning paradigm does not require labeled data? A) Supervised learning B) **Unsupervised learning** C) Reinforcement learning D) Semi-supervised learning	1	CO3	L2	Infosys Certification 2023
14	3	What is the space complexity of a feature set with n samples and m features? A) O(1) B) O(n)	1	CO3	L2	TCS NQT 2022

		C) **O(n*m)** D) O(m)				
15	3	Which of the following is a supervised learning task? A) **Classification** B) Data clustering C) Pattern discovery D) Anomaly detection	1	CO3	L3	GATE 2023
16	3	What is the time complexity of evaluating a model's performance? A) O(1) B) **O(n)** C) O(n^2) D) O(log n)	1	CO3	L1	TCS NQT 2022
17	3	Which of the following is a real-life application of unsupervised learning? A) **Customer segmentation** B) Email classification C) Predicting house prices D) Speech recognition	1	CO3	L2	Infosys Certification 2023
18	3	What is the main disadvantage of reinforcement learning? A) Requires large datasets B) **Slow convergence** C) High memory usage D) No need for data	1	CO3	L2	UGC-NET 2022
19	3	What is the space complexity of storing a reinforcement learning policy? A) O(1) B) O(n) C) O(n^2) D) **Depends on the policy**	1	CO3	L1	Infosys Certification 2023
20	3	Which process involves splitting data into subsets for training and testing? A) **Dataset division** B) Model training C) Feature engineering D) Evaluation	1	CO3	L2	GATE 2023
21	3	What is the purpose of a machine learning model?	5	CO3	L2	
22	3	Define stratified k-fold cross-validation with an example.	5	CO3	L2	
23	3	What is the time complexity of computing mean and variance?	5	CO3	L1	
24	3	Explain the concept of one-hot encoding in feature engineering.	5	CO3	L1	

25	3	What is the space complexity of a feature-engineered dataset?	5	CO3	L2	
26	3	Define a test set with an example.	5	CO3	L2	
27	3	What is the purpose of data quality in machine learning?	5	CO3	L3	
28	3	Explain the concept of leave-one-out cross-validation.	5	CO3	L2	
29	3	What is the time complexity of feature selection?	5	CO3	L2	
30	3	Define a validation set with an example.	5	CO3	L3	
31	3	Define the K-Nearest Neighbor algorithm.	5	CO3	L2	
32	3	What is the time complexity of training a linear regression model?	5	CO3	L1	
33	3	Explain the concept of a confusion matrix.	5	CO3	L3	
34	3	What is the purpose of the SSE metric in regression?	5	CO3	L3	
35	3	Define logistic regression with an example.	5	CO3	L2	
36	3	Discuss the time and space complexity of Support Vector Machines.	10	CO3	L4	
37	3	Describe the implementation of a confusion matrix for classification evaluation.	10	CO3	L4	
38	3	Explain the importance of precision, recall, and F-Score in classification.	10	CO3	L5	
39	3	Discuss the applications of supervised learning in real-world scenarios.	10	CO3	L5	
40	3	Discuss the role of cross-validation in preventing overfitting.	10	CO3	L6	
S.No.	**Unit**	**Question**	**Marks**	**CO**	**BL**	**Asked in Competitive Exam**
UNIT-4						
1	4	Which of the following is a reinforcement learning component? A) **Reward function** B) Feature set C) Confusion matrix D) ROC curve	1	CO4	L1	TCS NQT 2022
2	4	What is the time complexity of training a reinforcement learning model?	1	CO4	L1	Infosys Certificat

		A) O(1) B) O(n) C) $O(n^2)$ D) **Depends on the environment**				ion 2023
3	4	Which of the following is a supervised learning application? A) **Predicting house prices** B) Data clustering C) Pattern discovery D) Anomaly detection	1	CO4	L2	UGC-NET 2022
4	4	What is the main advantage of reinforcement learning? A) **Handles dynamic environments** B) Fast computation C) Low memory usage D) No overfitting	1	CO4	L2	Infosys Certification 2023
5	4	Which of the following is an unsupervised learning task? A) **Data clustering** B) Classification C) Regression D) Speech recognition	1	CO4	L2	GATE 2023
6	4	What is the space complexity of a machine learning pipeline? A) O(1) B) O(n) C) O(n*m) D) **Depends on the pipeline**	1	CO4	L2	UGC-NET 2022
7	4	Which of the following is a real-life application of machine learning? A) **Fraud detection** B) Manual data entry C) Network routing D) Hardware design	1	CO4	L2	TCS NQT 2022
8	4	What is the time complexity of applying a machine learning model to n samples? A) O(1) B) **O(n)** C) $O(n^2)$ D) O(log n)	1	CO4	L3	Infosys Certification 2023
9	4	Which learning paradigm combines labeled and unlabeled data? A) **Semi-supervised learning** B) Supervised learning C) Unsupervised learning D) Reinforcement learning	1	CO4	L1	UGC-NET 2022
10	4	What is the main disadvantage of supervised learning? A) **Requires labeled data** B) Fast computation C) Low accuracy	1	CO4	L2	GATE 2023

		D) No validation				
11	4	Which of the following is a machine learning process? A) **Hyperparameter tuning** B) Data storage C) Hardware optimization D) Network configuration	1	CO4	L2	TCS NQT 2022
12	4	What is the space complexity of storing a dataset for training? A) O(1) B) **O(n*m)** C) O(n) D) O(m)	1	CO4	L1	UGC-NET 2022
13	4	Which of the following is a real-life application of reinforcement learning? A) **Autonomous driving** B) Email classification C) Predicting house prices D) Data clustering	1	CO4	L2	TCS NQT 2022
14	4	What is the time complexity of feature selection for n samples and m features? A) **O(n*m)** B) O(n) C) O(n^2) D) O(log n)	1	CO4	L2	Infosys Certification 2023
15	4	Which of the following is a supervised learning algorithm? A) **Linear Regression** B) K-Means C) Hierarchical clustering D) Gaussian Mixture Model	1	CO4	L2	GATE 2023
16	4	What is the main advantage of unsupervised learning? A) **No need for labeled data** B) Fast computation C) Low memory usage D) High accuracy	1	CO4	L2	Infosys Certification 2023
17	4	What is the space complexity of a trained supervised learning model? A) O(1) B) O(m) C) O(n*m) D) **Depends on the model**	1	CO4	L3	TCS NQT 2022
18	4	Which of the following is a machine learning application in healthcare? A) **Disease prediction** B) Manual data entry C) Network routing D) Hardware design	1	CO4	L2	UGC-NET 2022

19	4	What is the time complexity of hyperparameter tuning using grid search? A) O(1) B) O(n) C) O(n^2) D) **Depends on the grid**	1	CO4	L1	TCS NQT 2022
20	4	Which learning paradigm is best for sequential decision-making? A) **Reinforcement learning** B) Supervised learning C) Unsupervised learning D) Semi-supervised learning	1	CO4	L2	GATE 2023
21	4	What is the time complexity of predicting with K-Nearest Neighbor?	5	CO4	L1	
22	4	Explain the concept of precision in classification.	5	CO4	L2	
23	4	What is the space complexity of a linear regression model?	5	CO4	L2	
24	4	Define the Support Vector Machine algorithm.	5	CO4	L2	
25	4	What is the time complexity of training a Support Vector Machine?	5	CO4	L2	
26	4	Explain the concept of the F-Score metric.	5	CO4	L3	
27	4	What is the purpose of the ROC curve in classification?	5	CO4	L1	
28	4	What is the space complexity of a K-Nearest Neighbor model?	5	CO4	L2	
29	4	Define the R^2 metric in regression.	5	CO4	L1	
30	4	What is the time complexity of computing a confusion matrix?	5	CO4	L3	
31	4	Explain the concept of recall in classification.	5	CO4	L2	
32	4	What is the space complexity of a Support Vector Machine model?	5	CO4	L3	
33	4	Define a classification task with an example.	5	CO4	L3	
34	4	What is the time complexity of evaluating a regression model?	5	CO4	L4	
35	4	Explain the concept of a regression task with an example.	5	CO4	L3	
36	4	Describe the implementation of a Support Vector Machine	10	CO4	L4	

		with an example.				
37	4	Explain the role of the ROC curve in evaluating classification models.	10	CO4	L5	
38	4	Discuss the time and space complexity of K-Nearest Neighbor.	10	CO4	L5	
39	4	Describe the implementation of grid search for hyperparameter tuning.	10	CO4	L5	
40	4	Explain the role of the MME metric in regression evaluation.	10	CO4	L6	
S.No.	**Unit**	**Question**	**Marks**	**CO**	**BL**	**Asked in Competitive Exam**
		UNIT-5				
1	5	Which algorithm is used for classification in supervised learning? A) **K-Nearest Neighbor** B) K-Means C) Hierarchical clustering D) Gaussian Mixture Model	1	CO5	L2	Infosys Certification 2023
2	5	What is the time complexity of training a linear regression model with n samples? A) **O(n)** B) $O(n^2)$ C) $O(n^3)$ D) O(log n)	1	CO5	L2	TCS NQT 2022
3	5	Which evaluation metric measures the squared error in regression? A) **SSE** B) Precision C) Recall D) F-Score	1	CO5	L2	UGC-NET 2022
4	5	What is the purpose of a confusion matrix in classification? A) **Evaluate model performance** B) Train the model C) Reduce features D) Store data	1	CO5	L2	TCS NQT 2022
5	5	Which algorithm is used for regression tasks? A) **Linear Regression** B) K-Means C) Hierarchical clustering D) Gaussian Mixture Model	1	CO5	L2	GATE 2023

6	5	What is the time complexity of predicting with K-Nearest Neighbor for n samples? A) O(1) B) **O(n)** C) $O(n^2)$ D) O(log n)	1	CO5	L3	UGC-NET 2022
7	5	Which metric evaluates the proportion of correct positive predictions? A) **Precision** B) Recall C) F-Score D) SSE	1	CO5	L2	Infosys Certification 2023
8	5	What is the main advantage of logistic regression? A) **Handles binary classification** B) Fast computation C) No overfitting D) Low memory usage	1	CO5	L2	TCS NQT 2022
9	5	What is the space complexity of a linear regression model with m features? A) O(1) B) **O(m)** C) O(n*m) D) O(n)	1	CO5	L2	UGC-NET 2022
10	5	Which algorithm is used for binary classification? A**) Support Vector Machine** B) K-Means C) Hierarchical clustering D) Gaussian Mixture Model	1	CO5	L1	GATE 2023
11	5	What is the time complexity of training a Support Vector Machine with n samples? A) O(n) B) **$O(n^2)$** C) $O(n^3)$ D) O(log n)	1	CO5	L2	Infosys Certification 2023
12	5	Which metric combines precision and recall? A) **F-Score** B) SSE C) R^2 D) ROC Curve	1	CO5	L2	UGC-NET 2022
13	5	What is the purpose of the ROC curve in classification? A) **Evaluate model performance** B) Train the model C) Reduce features D) Store data	1	CO5	L1	Infosys Certification 2023
14	5	What is the space complexity of a K-Nearest Neighbor	1	CO5	L2	TCS

		model? A) O(1) B) O(n) C) **O(n*m)** D) O(m)				NQT 2022
15	5	Which evaluation metric measures the goodness of fit in regression? A) **R^2** B) Precision C) Recall D) F-Score	1	CO5	L2	GATE 2023
16	5	What is the time complexity of computing a confusion matrix for n samples? A) O(1) B) **O(n)** C) $O(n^2)$ D) O(log n)	1	CO5	L3	TCS NQT 2022
17	5	Which algorithm is best for handling non-linear data? A) **Support Vector Machine** B) Linear Regression C) K-Nearest Neighbor D) Logistic Regression	1	CO5	L2	Infosys Certification 2023
18	5	What is the main disadvantage of K-Nearest Neighbor? A) **High memory usage** B) Fast computation C) No overfitting D) Low interpretability	1	CO5	L3	UGC-NET 2022
19	5	What is the space complexity of a Support Vector Machine model? A) O(1) B) **O(n)** C) $O(n^2)$ D) O(m)	1	CO5	L2	Infosys Certification 2023
20	5	Which metric is used to evaluate regression model errors? A) **SSE** B) Precision C) Recall D) F-Score	1	CO5	L2	GATE 2023
21	5	What is the purpose of the MME metric in regression?	5	CO5	L2	
22	5	Explain the concept of hyperparameter tuning in SVM.	5	CO5	L3	
23	5	What is the time complexity of predicting with a logistic regression model?	5	CO5	L3	
24	5	Define a soft-margin Support Vector Machine.	5	CO5	L2	

25	5	What is the space complexity of a confusion matrix?	5	CO5	L2	
26	5	Explain the concept of the AUC metric in classification.	5	CO5	L2	
27	5	What is the time complexity of computing recall?	5	CO5	L2	
28	5	Define a multi-class classification task with an example.	5	CO5	L3	
29	5	What is the space complexity of a logistic regression model?	5	CO5	L2	
30	5	Explain the concept of grid search in supervised learning.	5	CO5	L2	
31	5	What is the time complexity of training a K-Nearest Neighbor model?	5	CO5	L3	
32	5	Define the precision-recall curve with an example.	5	CO5	L3	
33	5	What is the space complexity of storing model predictions?	5	CO5	L2	
34	5	Explain the concept of a binary classification task.	5	CO5	L3	
35	5	What is the time complexity of computing the R^2 metric?	5	CO5	L2	
36	5	Describe the applications of regression models in real-world scenarios.	10	CO5	L4	
37	5	Explain the implementation of a soft-margin SVM with an example.	10	CO5	L5	
38	5	Discuss the role of the AUC metric in evaluating classifier performance.	10	CO5	L5	
39	5	Explain the implementation of K-Means clustering with a real-world example.	10	CO5	L5	
40	5	What is the purpose of the MME metric in regression?	10	CO5	L6	

Signature of Faculty/ Course Coordinator **Dean / Head of Department**